Reg. No									

St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore–607001

B. Sc DEGREE EXAMINATION - APRIL 2021

ASCS402T- STATISTICAL METHODS FOR COMPUTER APPLICATIONS – II

Time : 3 hrs **Max Marks :75**

SECTION - A (10x2=20)

Answer ALL the questions

1. What is meant by conditional probability?
2. Define mutually exclusive events in probability.
3. State any two property of mathematical expectations.
4. What is random variable?
5. State the condition under which the binomial distribution tends to the Poisson distribution.
6. Define standard normal variate.
7. Define level of significance.
8. What is mean by critical region?
9. Explain local control.
10. Explain the Randomized Design.

SECTION - B (5×5=25)

Answer ALL the questions

11. a) Two persons A and B are toss a coin alternatively. The person who gets head first wins the game. Find the chances of A win the game.

(Or)

b) A bag contains 4 white and 6 black balls. Two balls are drawn at random. What is the probability that
   (i) Both are white
   (ii) Both are black
   (iii) Both are different colors.

12. a) Find the mean of X if $f(x) = \begin{cases} \frac{3}{(x+1)4} & , 0 < x < \infty \\ 0 & , otherwise \end{cases}$

(Or)

b) A coin is tossed until a head appears. What is the expectation of the number of tosses?

13. a) For a binomial distribution the mean is 4 and variance is 2. Find the probability of getting
(i) atleast 2 successes
(ii) at most 2 successes

(Or)

b) What are the characteristics of normal distribution and state any 5.

14. a) Define Type I error and Type II error.

(Or)

b) Write the procedure for the testing of hypothesis.

15. a) Write the procedure for work out the CRD.

(Or)

b) What are the basic principles of Design of Experiment and Explain.

SECTION - C (3×10=30)

Answer any THREE Questions

16. A,B, and C wash 50%, 30% and 20% of the cars in a service station respectively. They fail to clean the glass in 8% , 12% and 5% of the cars respectively. The glass of a washed car is checked. What is the probability that the glass has been cleaned?

17. State and prove Chebychev's inequality.

18. The customer account of a certain department store have an average balance of Rs .120 and standard deviation of Rs.40 . Assume that the account balance are normally distributed, find

(i) What proportion of accounts is over Rs.150?
(ii) What proportion of accounts is between Rs.100 and Rs.150?
(iii) What proportion of accounts is between Rs.60 and Rs.90?

19. In a sample of 600 men and from a certain city, 450 men are found to be smokers. In a sample of 900 from another city, 450 men are found to be smokers. Do the data indicate that the two cities are significantly different with respect to prevalence of smoking habit among men?

20. Carry out a complete ANOVA for the following data and comment on the results.

A	20	28	42	36	29	40	28
B	18	22	29	43	40	28	
C	42	40	36	28	12	19	20
D	18	24	42	40	52	46	
E	40	20	12	19	17		

Test Whether the outputs of the 5 Processes are equal or not.

****